

Hierarchical Clustering

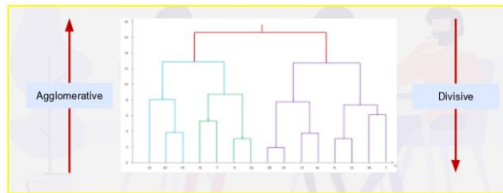
Intuition, dendrograms, linkage, and the mathematics of greedy merges

April 2, 2026

Why Hierarchical Clustering?

- ▶ Hierarchical clustering is **unsupervised** and **connectivity-based**.
- ▶ It builds a **nested family of clusters**, not just one partition.
- ▶ We do *not* need to pick one value of K up front.
- ▶ The raw ingredients are observations plus a dissimilarity:

observations + a dissimilarity $d(x_i, x_j)$.
- ▶ Nearby points merge low in the tree; distant groups only merge high up.



Agglomerative Versus Divisive

Agglomerative

- ▶ Bottom-up
- ▶ Start with n singleton clusters
- ▶ Repeatedly merge the two closest clusters
- ▶ Most common form in practice

Divisive

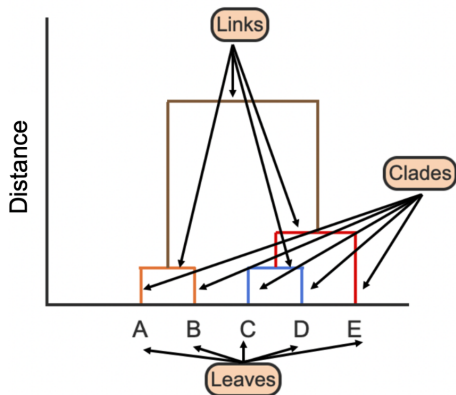
- ▶ Top-down
- ▶ Start with one large cluster containing every point
- ▶ Repeatedly split a cluster into smaller pieces
- ▶ Conceptually natural, but less common in standard software

singletons $\longrightarrow$ larger and larger groups

one big cluster $\longrightarrow$ finer and finer groups

- ▶ In this deck, “hierarchical clustering” means **agglomerative hierarchical clustering** unless stated otherwise.

How to Read a Dendrogram



Adapted from
refs/10-hierarchical-clustering.pptx

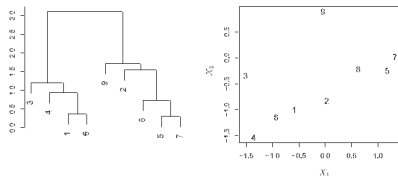
- ▶ Each **leaf** is one observation.
- ▶ A horizontal **link** marks the moment when two clusters merge.
- ▶ The **height** of that link is the dissimilarity at which the merge happened.
- ▶ Clusters that join low in the tree are more similar than clusters that join high in the tree.

Interpretation

The vertical axis matters much more than the horizontal ordering of the leaves.

Interpreting Dendrograms Carefully

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Adapted from

quizzes/book-snips/ISLP-12.4-Clustering.pdf

- ▶ The **height** of a merge is meaningful; the left-to-right ordering of leaves usually is not.
- ▶ A point may sit horizontally near another point in the dendrogram but still merge with it at a much larger height.
- ▶ Always connect the dendrogram back to the original feature space or distance matrix before making a story out of it.

Rule of thumb

Read dendrograms vertically, not horizontally.

Agglomerative Clustering Algorithm

Start with a pairwise dissimilarity matrix

$$D_{ij}^{(0)} = d(x_i, x_j), \quad D_{ii}^{(0)} = 0.$$

Greedy merge rule

At step t , choose

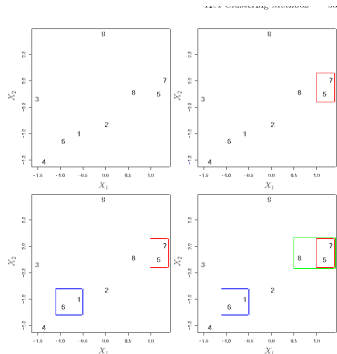
$$(A_t, B_t) = \arg \min_{A \neq B} d_{\text{link}}(A, B),$$

merge them, and record the merge height

$$h_t = d_{\text{link}}(A_t, B_t).$$

Initialize with n singleton clusters, repeatedly merge and update, and stop when one cluster remains.

The method is **greedy**: once two clusters merge, that choice cannot be undone.



The Proximity Matrix Drives the Merges

For observations $x_1, \dots, x_n$, start with the pairwise dissimilarity matrix

$$D = (d_{ij})_{i,j=1}^n, \quad d_{ij} = d(x_i, x_j).$$

$$D = \begin{bmatrix} 0 & d_{12} & \cdots & d_{1n} \\ d_{21} & 0 & \cdots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \cdots & 0 \end{bmatrix}$$

- ▶ It is symmetric when $d(x_i, x_j) = d(x_j, x_i)$.
- ▶ Diagonal entries are 0 because each point has zero distance from itself.

What changes after a merge?

If A and B merge into $A \cup B$, then every remaining cluster C needs a new distance

$$d_{\text{link}}(A \cup B, C).$$

- ▶ After each merge, two rows/columns are replaced by one for the merged cluster.
- ▶ Single linkage uses the closest cross-cluster pair.
- ▶ Complete linkage uses the farthest cross-cluster pair.
- ▶ Average linkage uses the mean cross-cluster distance.

The Key Mathematical Choice: Linkage

For clusters A and B , the linkage function defines their distance:

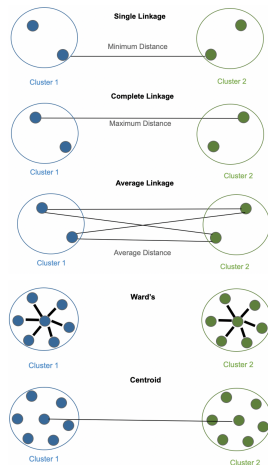
$$d_{\text{single}}(A, B) = \min_{i \in A, j \in B} d(x_i, x_j)$$

$$d_{\text{complete}}(A, B) = \max_{i \in A, j \in B} d(x_i, x_j)$$

$$d_{\text{average}}(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} d(x_i, x_j)$$

$$d_{\text{centroid}}(A, B) = \|\bar{x}_A - \bar{x}_B\|_2$$

$$d_{\text{Ward}}(A, B) = \text{ESS}(A \cup B) - \text{ESS}(A) - \text{ESS}(B)$$



Ward's Linkage: The Intuition

Ward's method does **not** ask which two clusters have the smallest point-to-point distance. Instead, it asks:

Guiding question

If we merge clusters A and B , how much more **spread out** will the data become?

- ▶ A good merge should keep the new cluster **tight and compact**.
- ▶ A bad merge would glue together groups that create a large, diffuse cloud.
- ▶ So Ward picks the pair whose merge causes the **smallest increase** in within-cluster variation.

Compared with other linkage rules

- ▶ **Single**: nearest pair.
- ▶ **Complete**: farthest pair.
- ▶ **Average**: mean pairwise distance.
- ▶ **Ward**: the **whole cluster geometry**.

Practical effect

Ward often avoids chaining and tends to prefer rounder, more balanced clusters.

A Tiny Worked Example of Linkage Values

Suppose we have two clusters on the real line:

$$A = \{0, 2\}, \quad B = \{5, 6\}.$$

The cross-cluster distances are

$$5, 6, 3, 4.$$

Linkage values

$$d_{\text{single}}(A, B) = 3$$

$$d_{\text{complete}}(A, B) = 6$$

$$d_{\text{average}}(A, B) = \frac{5 + 6 + 3 + 4}{4} = 4.5$$

$$d_{\text{centroid}}(A, B) = |1 - 5.5| = 4.5$$

For Ward linkage,

$$\bar{x}_A = 1, \quad \bar{x}_B = 5.5.$$

Using the Euclidean simplification,

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\bar{x}_A - \bar{x}_B\|_2^2 = \frac{2 \cdot 2}{4} (4.5)^2 = 20.25.$$

Takeaway

The *same two clusters* can look close or far apart depending on the linkage rule, which is why linkage choice strongly affects the dendrogram.

Distance Choice Matters Too

If scales differ, standardize first

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$$

- ▶ Otherwise, variables with large units can dominate the distance calculation.
- ▶ A spending variable measured in dollars can overwhelm an age variable measured in years.
- ▶ Standardization makes each feature contribute on a comparable scale.

Pick the dissimilarity to match the question

$$d_{\text{Euclidean}}(x_i, x_j) = \|x_i - x_j\|_2$$

$$d_{\text{corr}}(x_i, x_j) = 1 - \text{corr}(x_i, x_j)$$

- ▶ Euclidean distance emphasizes absolute magnitude.
- ▶ Correlation-based distance emphasizes **shape** or profile similarity.
- ▶ The right distance depends on the scientific goal, not just convenience.

Ward Linkage: The Objective Function

Within-cluster error

$$\text{ESS}(C) = \sum_{i \in C} \|x_i - \bar{x}_C\|_2^2$$
$$\bar{x}_C = \frac{1}{|C|} \sum_{i \in C} x_i$$

Ward linkage merges the pair that yields the **smallest increase** in total within-cluster sum of squares:

$$\Delta(A, B) = \text{ESS}(A \cup B) - \text{ESS}(A) - \text{ESS}(B).$$

For Euclidean distance, this simplifies to

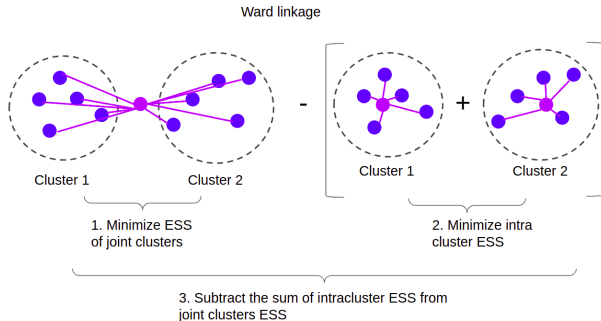
$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\bar{x}_A - \bar{x}_B\|_2^2.$$

Small centroid gaps and balanced sizes make a merge attractive.

Interpretation

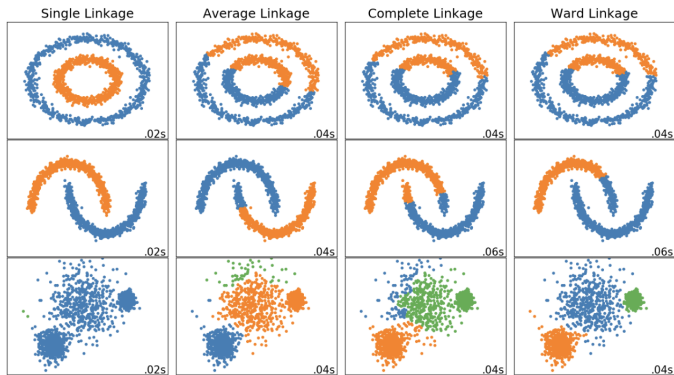
- ▶ Ward is a greedy least-squares method.
- ▶ It often prefers compact, roughly spherical clusters.
- ▶ This makes it feel like a hierarchical cousin of *k*-means.

Ward Linkage: Visual Intuition



- ▶ First compute the error of the merged cluster.
- ▶ Then subtract the errors of the two original clusters.
- ▶ The best merge is the one with the **smallest increase** in total within-cluster variation.

Different Linkages Can Tell Very Different Stories



- ▶ **Single linkage** can recover non-convex shapes, but it is prone to *chaining*.
- ▶ **Complete** and **average linkage** usually create more balanced dendrograms.
- ▶ **Ward linkage** often behaves like a hierarchical cousin of *k*-means: it likes compact groups.

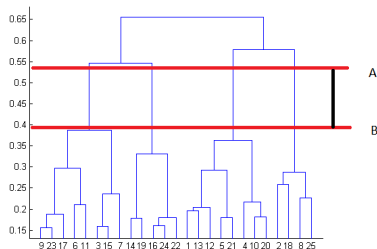
Cutting the Dendrogram Gives a Clustering

Choose a height h and remove every merge above that level. The remaining connected components define the clusters:

$\mathcal{C}_h = \{\text{connected components below height } h\}$.

- ▶ Higher cut $\Rightarrow$ fewer clusters.
- ▶ Lower cut $\Rightarrow$ more clusters.
- ▶ In practice, we look for a large vertical gap before making the cut.

K-means chooses K first; hierarchical clustering builds the whole tree first and chooses K by the cut.



Nested clusters are an assumption

Cutting a dendrogram at height h_1 and then at a lower height h_2 always produces nested groups.

- ▶ That nested structure is a property of the algorithm, not proof that the data are truly hierarchical.
- ▶ In some data sets, the best 2-group split and best 3-group split may not be consistent with one another.

Practical consequence

Hierarchical clustering can sometimes give worse results than methods like k -means for a fixed number of clusters.

- ▶ Greedy early merges cannot be repaired later.
- ▶ Compare several linkage and scaling choices rather than trusting a single tree blindly.

A Tempered Approach to Interpreting Clusters

- ▶ Small choices can have big consequences: scaling, distance measure, and linkage can all change the result.
- ▶ There is rarely a single “correct” clustering waiting to be discovered automatically.
- ▶ A good workflow is to compare several reasonable specifications and look for patterns that recur.
- ▶ If the same clusters appear across subsets of the data, that gives more confidence that the structure is robust.
- ▶ Treat clustering as an **exploratory tool**: a starting point for scientific interpretation, not the last word.

Practical habit

Use clustering to generate hypotheses and summaries, then validate those ideas with domain knowledge or new data whenever possible.

Strengths and Limitations

Strengths

- ▶ No need to choose one partition in advance
- ▶ Works with many dissimilarity measures, not just Euclidean distance
- ▶ Dendrograms are often easy to explain to humans
- ▶ Useful for exploratory analysis and small-to-medium data sets

Limitations

- ▶ Distances can change dramatically if variables are not scaled
- ▶ Early greedy merges cannot be undone later
- ▶ Results depend strongly on the choice of linkage and distance metric
- ▶ Pairwise distances require about $O(n^2)$ memory

memory $\approx O(n^2)$, time typically ranges from $O(n^2 \log n)$ to $O(n^3)$.

Takeaways

- ▶ Hierarchical clustering builds a **tree of nested clusters**, not just one partition.
- ▶ Agglomerative clustering is a greedy bottom-up procedure that repeatedly merges the closest clusters.
- ▶ The **linkage function** is the main mathematical choice because it defines what “closest” means for clusters.
- ▶ Distances, scaling, and linkage choices all shape the final dendrogram, so interpretation should be cautious.
- ▶ Ward linkage minimizes the increase in within-cluster sum of squares, linking hierarchical clustering to least-squares ideas.
- ▶ A dendrogram becomes a concrete clustering only after we choose a **cut height**.